Yissum Research Intelligence Report — Healthcare — July 2026

17 high-potential paper(s) · Healthcare · Monthly · July 2026 · Generated July 22, 2026

Hebrew University researchers are pioneering advanced precision medicine approaches across several high-impact fields. This includes developing novel therapeutics for challenging diseases like Multiple Sclerosis, chronic pain, and various cancers, alongside sophisticated diagnostic platforms and AI-powered tools that significantly accelerate drug discovery. Furthermore, the university is advancing public health solutions through enhanced infection control technologies. These breakthroughs represent significant commercial opportunities in rapidly evolving biotechnology and healthcare markets, addressing critical unmet needs and streamlining R&D pipelines.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Avi Priel** — Drug Discovery, Neuroscience, Clinical
- **Oren Zimhony** — Medical Device, Materials, Clinical
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Batsheva Kerem** — Drug Discovery, Genomics, Clinical
- **Rachel Nechushtai** — Drug Discovery, Proteomics, Clinical
- **Dmitry Tsvelikhovsky** — Drug Discovery, Materials, Synthetic Biology
- **Dina Schneidman-Duhovny** — Drug Discovery, Immunology, Software/AI, Synthetic Biology
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Ayelet N Landau** — Diagnostics, Medical Device, Software/AI, Neuroscience
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology
- **Ofer Mandelboim** — Immunology, Drug Discovery, Clinical
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Yuval Ramot** — Software/AI, Immunology, Clinical, Drug Discovery
- **Ora Schueler-Furman** — Drug Discovery, Software/AI, Proteomics
- **Haitham Amal** — Drug Discovery, Neuroscience, Clinical

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Off-the-shelf stem cell therapy shows promise for remyelination in MS and neurodegeneration."

Main researcher: Tamir Ben-Hur · **tamir@hadassah.org.il.**

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

WHY NOW

This addresses a critical unmet need in neurodegenerative diseases, with a clinical-grade therapy offering a scalable approach to remyelination. The growing market for advanced cell therapies provides a clear pathway for rapid development and investment.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty 6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential 8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size 9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness 5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength 8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

View in Dashboard Open Paper

2. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells. 35/50

"Next-gen single-cell RNA sequencing platform reveals hidden disease drivers for precision medicine."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

WHY NOW

The rapidly expanding precision medicine market demands more comprehensive and cost-effective genomic tools. This platform's ability to uncover new disease mechanisms offers a competitive edge in identifying novel therapeutic targets and advanced diagnostics.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty 8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential 8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size 8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness 4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength 7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

View in Dashboard Open Paper

3. Carbamate-based cannabinoids uncover a conserved lipid-sensing pocket in nociceptive TRPA1 and TRPV1 channels. 33/50

"Novel carbamate cannabinoids offer potent, non-opioid analgesia for chronic pain management."

Main researcher: Avi Priel · avi.priel@mail.huji.ac.il.

The Institute for Drug Research (IDR), Department of Pharmacology, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Israel. Electronic address: avi.priel@mail.huji.ac.il.

Published in Pharmacological research · 2026-06-24

WHY NOW

The global opioid crisis and urgent demand for safer pain treatments create a massive market need for effective non-opioid analgesics. These engineered compounds provide a structural blueprint for a high-value drug development pipeline.

COMMERCIAL ANGLE

The development of these high-potency carbamate derivatives provides a structural roadmap for designing a new class of non-opioid analgesics for chronic pain management.

COMMERCIALISATION METRICS

Novelty 7/10

The research demonstrates significant chemical innovation by using rational design to transform known scaffolds into potent dual agonists, revealing a previously unknown conserved lipid-sensing mechanism. While it offers high target-validation value and potential for new drug classes, it builds incrementally on established phytocannabinoid chemistry rather than introducing an entirely new paradigm.

Commercial Potential 7/10

The research demonstrates successful rational design of small-molecule derivatives with enhanced potency and selectivity, representing a clear path toward a novel non-opioid analgesic lead series. However, the transition from structural discovery to a proprietary drug candidate requires significant validation of pharmacokinetic profiles and safety to reach high-tier commercial readiness.

Market Size 9/10

The research targets TRPA1 and TRPV1, two of the most high-value pathways in the multi-billion dollar non-opioid analgesic market. By developing potent dual-agonists, this work addresses a massive unmet medical need for chronic and acute pain management with significant commercial scalability.

Tech Readiness 3/10

The research is currently at the basic discovery/proof-of-concept stage, utilizing in vitro heterologous expression systems to validate molecular binding and potency. It lacks in vivo animal models, pharmacokinetic data, or toxicity profiling required to advance toward preclinical development.

IP Strength 7/10

The research demonstrates successful rational ligand design to create novel, highly potent carbamate-functionalized derivatives with improved selectivity, providing a clear pathway for composition-of-matter patent claims. While the chemical scaffolds are derivative of known phytocannabinoids, the specific structural modifications and the discovery of a conserved binding pocket offer significant defensibility for new non-opioid analgesic candidates.

View in Dashboard Open Paper

4. The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms. **33/50**

"Proprietary surface coating dramatically reduces multidrug-resistant bacteria in healthcare environments."

Main researcher: Oren Zimhony · Oren_Z@clalit.org.il.

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Published in Antimicrobial resistance and infection control · 2026-06

WHY NOW

Hospitals face immense pressure to combat healthcare-associated infections and antibiotic resistance. This scalable, durable coating offers an immediate, cost-effective solution to enhance existing disinfection protocols, tapping into a large infection control market.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

COMMERCIALISATION METRICS

Novelty **5/10**

The research presents an incremental improvement on existing disinfection protocols by using a proprietary coating to extend the efficacy of standard chlorine-based agents. While clinically useful, the innovation is an additive formulation rather than a groundbreaking paradigm shift in antimicrobial science.

Commercial Potential **8/10**

The technology demonstrates a clear, additive value proposition to existing hospital disinfection protocols and uses a validated proof-of-concept to address high-value clinical needs (MDRO reduction). The 'proprietary formulation' suggests protectable IP that can be easily commercialized as a specialized chemical adjunct for the healthcare sector.

Market Size **6/10**

The technology addresses a high-value niche in infection control (MDRO isolation rooms), but its application as an adjunct to existing protocols suggests a specialized additive market rather than a broad-scale replacement for global disinfection commodities.

Tech Readiness **7/10**

The technology has moved beyond lab-scale validation into a controlled quasi-experimental clinical setting with real-world implementation in hospital isolation rooms. It is currently at the stage of clinical validation/pilot testing, positioned just before full-scale commercial rollout or large-scale multi-center trials.

IP Strength **7/10**

The use of a proprietary formulation (OxiLast™) to stabilize existing disinfectants suggests a protectable chemical composition or method-of-use patent. However, the efficacy relies on an additive application to standard protocols, which may face moderate defensibility challenges against generic formulations or easier-to-replicate chemical stabilizers.

View in Dashboard Open Paper

5. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Self-delivering genetic analogues enable facile intracellular targeting for next-gen therapeutics."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

WHY NOW

Overcoming delivery barriers is a key challenge for genetic therapies. This platform streamlines cellular uptake, opening new avenues for developing high-stability, cost-effective nucleic acid-based drugs without complex delivery systems.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty 6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential 8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size 9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness 3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength 7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

View in Dashboard Open Paper

6. SPL84 and trikafta® exhibit comparable and additive effects in patient-derived HBE cells carrying the 3849+10kb C→T/F508del CFTR variants. 31/50

"Combination therapy improves outcomes for cystic fibrosis patients poorly responding to current drugs."

Main researcher: Batsheva Kerem · batshevak@savion.huji.ac.il.

SpliSense, Haddasah Ein Kerem, Jerusalem, Israel; Department of Genetics, The Hebrew University, Jerusalem, Israel. Electronic address: batshevak@savion.huji.ac.il.

Published in Journal of cystic fibrosis : official journal of the European Cystic Fibrosis Society · 2026-06-26

WHY NOW

This research targets an underserved segment of CF patients who respond sub-optimally to current treatments, representing a valuable precision medicine opportunity. Combining with existing drugs offers a fast-track to market via repurposing and add-on indications.

COMMERCIAL ANGLE

Developing a combination therapy or a precision medicine regimen that pairs existing CFTR modulators with targeted antisense oligonucleotides to capture the underserved segment of patients with modest responses to current treatments.

COMMERCIALISATION METRICS

Novelty 6/10

The research is an incremental optimization study focused on combination therapy rather than a fundamental discovery of a new biological mechanism or platform. While the additive effect of an ASO with an existing standard of care is clinically relevant, it represents an evolutionary application of known splicing modulation and CFTR corrector technologies.

Commercial Potential 8/10

The study demonstrates a clear additive therapeutic strategy for a specific, high-value patient population, providing a strong de-risking signal for a combination product approach. The focus on an inhaled ASO targeting a known clinical gap in current standard-of-care (Trikafta) represents a highly viable and direct path to clinical application.

Market Size 4/10

The target market is limited to a highly specific genetic sub-population (3849+10kb variant) within the already established cystic fibrosis patient base. While clinically valuable for non-responders, the addressable demand is niche and constrained by the prevalence of this specific splicing mutation.

Tech Readiness 6/10

The asset has already completed a Phase 2a clinical trial, indicating significant de-risking; however, the current paper focuses on ex vivo validation in patient-derived cells rather than advancing the clinical trial phase.

IP Strength 7/10

The innovation utilizes a specific antisense oligonucleotide (ASO) sequence targeting a known splicing defect, which is highly patentable and provides a clear pathway for composition-of-matter protection. However, the defensibility is slightly tempered by the fact that it is a combination therapy with an existing standard-of-care, which may face competition from other splicing modulators.

View in Dashboard Open Paper

7. Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein. **31/50**

"Targeted peptide therapeutic shows high efficacy and low toxicity for advanced ovarian cancer."

Main researcher: Rachel Nechushtai · rachel@mail.huji.ac.il.

Institute of Life Science, Faculty of Science and Mathematics, The Hebrew University of Jerusalem, Edmond J. Safra Campus at Givat Ram, Jerusalem 9190401, Israel. Electronic address: rachel@mail.huji.ac.il.

Published in Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie · 2026 Jul 1

WHY NOW

Advanced ovarian cancer has high relapse rates and poor survival, creating a significant unmet need for novel treatments. This first-in-class peptide offers a potentially safer and more effective therapeutic option for a high-value oncology indication.

COMMERCIAL ANGLE

Development of a first-in-class targeted peptide therapeutic to address the high relapse rates and low survival outcomes of advanced ovarian cancer.

COMMERCIALISATION METRICS

Novelty **6/10**

The use of a protein-derived peptide for selective cancer cytotoxicity is a known strategy; however, targeting the NAF-1/CISD2 pathway specifically for ovarian cancer represents a novel application of this protein's biology.

Commercial Potential **7/10**

The peptide addresses a high-unmet-need market (refractory ovarian cancer) with demonstrated selectivity and in vivo efficacy, offering a clear path to a therapeutic product. However, the score is tempered by the typical pharmacological challenges of peptides, such as stability and delivery, which are not addressed in the abstract.

Market Size **8/10**

Ovarian cancer represents a high-value target market due to a massive global incidence rate and a critical lack of effective treatments for relapsed or metastatic cases. The addressable demand is significant given the high relapse rate (80%) and poor 5-year survival outcomes.

Tech Readiness **3/10**

The technology is at a basic proof-of-concept stage, having demonstrated efficacy in vitro and in xenograft mouse models, which corresponds to early preclinical development.

IP Strength **7/10**

The innovation centers on a specific peptide sequence derived from a human protein (NAF-1/CISD2), which is typically highly patentable as a composition of matter. While the specificity for ovarian cancer provides strong defensibility, the score is tempered by the inherent vulnerability of peptides to proteolysis and potential ease of sequence modification by competitors.

View in Dashboard Open Paper

8. Proline-promoted electrophilic dichloromethylation of α-enaminones via stereochemically gated resolution. 30/50

"Sustainable, metal-free chemical synthesis platform streamlines pharmaceutical and advanced materials manufacturing."

Main researcher: Dmitry Tsvelikhovsky · dmitryt@ekmd.huji.ac.il.

The Institute for Drug Research, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, 9112102, Israel. dmitryt@ekmd.huji.ac.il.

Published in Nature communications · 2026-04

WHY NOW

The pharmaceutical industry constantly seeks more efficient, environmentally friendly, and cost-effective synthesis methods. This versatile platform offers a competitive advantage in producing complex chemical building blocks for high-value products.

COMMERCIAL ANGLE

This platform provides a more efficient and sustainable way to synthesize complex building blocks used in pharmaceutical manufacturing and advanced materials science.

COMMERCIALISATION METRICS

Novelty 8/10

The research introduces a novel closed-shell, metal-free mechanism for polyhalomethylation that shifts away from traditional radical/carbene pathways. The discovery of topology-driven divergence—where ring size dictates either functionalization or spontaneous cascade aromatization—represents a significant, programmable platform for synthetic methodology.

Commercial Potential 6/10

The paper describes a highly versatile, metal-free synthetic platform that could streamline the production of complex building blocks for medicinal chemistry. However, its commercial value is currently limited to a specialized tool for process chemists rather than a stand-alone product or a transformative platform with broad industrial scalability.

Market Size 6/10

The platform offers valuable versatility for medicinal chemistry and fine chemical synthesis by enabling novel functionalization of building blocks, but its market size is constrained to the niche specialty chemical and R&D reagent sectors rather than a large-scale therapeutic or industrial commodity market.

Tech Readiness 3/10

This is fundamental discovery research focused on mechanistic understanding and new synthetic methodology at the academic lab stage. While it demonstrates a novel chemical platform, it lacks the process optimization, scalability testing, or applied proof-of-concept required for higher TRL scores.

IP Strength 7/10

The discovery of a unique, metal-free, closed-shell mechanistic class offers strong potential for broad composition-of-matter and process patents in medicinal chemistry. However, the utility is currently limited to fundamental synthetic methodology, which may face challenges in demonstrating high-value industrial exclusivity compared to targeted therapeutic or material platforms.

View in Dashboard Open Paper

9. Progress in structure prediction and design of adaptive immune receptors. 30/50

"AI platform accelerates precision design of synthetic immune receptors for personalized cancer therapies."

Main researcher: Dina Schneidman-Duhovny · dina.schneidman@mail.huji.ac.il.

The Rachel and Selim Benin School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel. Electronic address: dina.schneidman@mail.huji.ac.il.

Published in Current opinion in structural biology · 2026 Jul 10

WHY NOW

The burgeoning field of cancer immunotherapy demands sophisticated tools for designing highly effective and personalized treatments. This AI-driven platform addresses a critical bottleneck in developing next-generation antibodies and T-cell therapies.

COMMERCIAL ANGLE

Developing AI-driven platforms for the precision design of synthetic antibodies and T-cell receptors for personalized cancer immunotherapies.

COMMERCIALISATION METRICS

Novelty 4/10

Based on the title, this is a review paper summarizing 'progress' rather than a primary research article presenting a novel method or discovery. It synthesizes existing advancements in the field rather than introducing groundbreaking innovation.

Commercial Potential 8/10

Structure prediction for adaptive immune receptors is a platform-enabling capability that directly accelerates the design of high-affinity therapeutic antibodies and TCR-T cell therapies, representing a massive commercial market in precision medicine.

Market Size 9/10

The ability to predict and design adaptive immune receptors is a platform-enabling technology with massive addressable demand across biologics, personalized cancer vaccines, and autoimmune therapies.

Tech Readiness 3/10

The title indicates a review or research progress report on structure prediction and design, which is fundamentally a computational/foundational stage of development. It focuses on modeling and design principles rather than a validated clinical candidate or a commercialized diagnostic platform.

IP Strength 6/10

The work focuses on structure prediction and design for adaptive immune receptors, which provides a strong foundation for platform-based IP in biologics; however, without a specific novel algorithm or proprietary dataset mentioned, it likely represents incremental progress in an open-science field.

View in Dashboard Open Paper

10. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling by nitric oxide inhibition in the Shank3 mouse model of autism. 28/50

"Identified molecular targets enable precision pharmacological interventions for autism spectrum disorders."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Haitham.amal@mail.huji.ac.il.; The Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.

Published in Scientific reports · 2026-06

WHY NOW

Autism spectrum disorders represent a significant unmet medical need, with a growing demand for targeted therapies. Identifying specific molecular pathways in genetic models offers a clear roadmap for developing precision drugs.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty 6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential 6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size 9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness 3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength 4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

View in Dashboard Open Paper

11. Universal rhythmic architecture uncovers two modes of neural dynamics. 28/50

"Universal neural dynamics framework enables advanced neurodiagnostic software and brain-computer interfaces."

Main researcher: Ayelet N Landau

Department of Psychology, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Cognitive and Brain Sciences, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Experimental Psychology, University College London (UCL), London, WC1H 0AP, UK.

Published in Nature communications · 2026-05

WHY NOW

The increasing demand for personalized neurodiagnostics and rapid advancements in brain-computer interfaces (BCIs) create a fertile market. This foundational understanding allows for more precise and adaptable neurotechnologies.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

COMMERCIALISATION METRICS

Novelty 8/10

The research shifts the fundamental paradigm of neural analysis from a frequency-only model to a two-dimensional architecture (frequency + rhythmicity), providing a potentially transformative framework for neurotechnological diagnostics. This provides a platform-enabling discovery that unifies disparate datasets across species and clinical states into a single organizing principle.

Commercial Potential 5/10

While the framework offers high potential for improving the precision of neurodiagnostic biomarkers and personalized EEG analysis, it remains a fundamental descriptive discovery rather than a specific device or therapeutic intervention. Its commercial value depends entirely on its integration into future clinical software or neurotechnology platforms.

Market Size 8/10

The findings provide a platform-enabling framework for universal neurophysiological analysis, addressing a massive market spanning clinical diagnostics, neuroprosthetics, and brain-computer interfaces across diverse patient populations.

Tech Readiness 3/10

The paper presents fundamental discovery-based neuroscience that provides a new theoretical framework for understanding neural dynamics. While it offers potential for future diagnostic biomarkers, it is currently at a basic research stage without a defined prototype, hardware integration, or clinical validation pipeline.

IP Strength 4/10

The work describes a scientific framework and a descriptive method for analyzing existing neural data rather than a novel hardware device or a proprietary biochemical entity. While the 'rhythmicity measure' could potentially be patented as a diagnostic algorithm, the primary output is foundational knowledge that is difficult to defend against competitors using alternative mathematical modeling.

View in Dashboard Open Paper

12. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI framework distills single-cell data, uncovering disease drivers and biomarkers for drug discovery."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. Electronic address: mor.nitzan@mail.huji.ac.il.

Published in Cell systems · 2026-05

WHY NOW

Pharmaceutical companies are heavily investing in single-cell genomics for biomarker discovery and target identification. This high-throughput AI software offers a crucial advantage in extracting actionable insights from noisy, high-dimensional datasets.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty 6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential 6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size 8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness 4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength 4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

View in Dashboard Open Paper

13. RadD from <i>Fusobacterium nucleatum</i> engages NKp46 to promote antitumor cytotoxicity. 28/50

"Bacterial protein activates NK cells, offering a novel mechanism to enhance anti-tumor immunotherapy."

Main researcher: Ofer Mandelboim

The Concern Foundation Laboratories at the Lautenberg Center for Immunology and Cancer Research, Institute for Medical Research Israel Canada (IMRIC), Hebrew University Hadassah Medical School, Jerusalem, Israel.

Published · 2026-05

WHY NOW

Natural killer (NK) cell therapies are a rapidly advancing area in oncology, with high potential. Discovering a novel, potent agonist provides a unique mechanism for developing next-generation immunotherapies to fight cancer.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

COMMERCIALISATION METRICS

Novelty 7/10

The discovery of a specific bacterial protein (RadD) acting as a direct ligand for the NKp46 receptor is highly novel and expands the known ligandome of NK cells. While it provides a strong foundation for targeted immunotherapies, it is an incremental discovery of a molecular interaction rather than a platform-level technological breakthrough.

Commercial Potential 6/10

The identification of RadD as a specific ligand for NKp46 provides a clear molecular target for developing NK cell agonists or mimics to enhance antitumor cytotoxicity. However, the commercial path is complicated by the dual nature of F. nucleatum in cancer progression and the need for precise delivery to avoid systemic inflammation.

Market Size 6/10

The target market encompasses oncology immunotherapy, specifically head and neck cancers, which is substantial; however, the current focus is on a highly specific host-microbe interaction rather than a broad platform application.

Tech Readiness 3/10

The research is at the fundamental discovery stage, establishing a biological mechanism through in vitro assays and murine models. While it identifies a novel target for immunotherapy, there are no lead candidates, pharmacokinetic data, or clinical validation steps presented.

IP Strength 6/10

The identification of a specific bacterial ligand (RadD) for a known human receptor (NKp46) provides a clear target for composition-of-matter patents on RadD-mimetics or agonists. However, since RadD is a naturally occurring bacterial protein, the IP strength depends on the ability to engineer novel derivatives or specific delivery vehicles rather than the discovery of the axis itself.

View in Dashboard Open Paper

14. Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2 27/50

"Small-molecule therapeutic targeting nitric oxide shows promise for genetically defined autism subtypes."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

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Published in Molecular autism · 2026-04

WHY NOW

Building on recent discoveries, this offers a concrete pharmacological strategy for a specific, high-need patient population in autism. The potential for a small-molecule drug provides a clear and attractive development path.

COMMERCIAL ANGLE

The findings support the development of small-molecule therapeutics targeting nitric oxide synthase as a precision medicine approach for genetically defined autism subtypes.

COMMERCIALISATION METRICS

Novelty 5/10

The study follows a logical progression from previously established findings (implicated NO signaling) to mechanistically characterizing those findings via proteomics. While it provides important biological validation, the innovation is incremental rather than a groundbreaking discovery of a new pathway or a novel therapeutic modality.

Commercial Potential 6/10

The study identifies a specific molecular target (nNOS inhibition) and a drug (7-NI) that shows functional efficacy in two different genetic models of ASD, providing a clear path toward drug repurposing or development. However, the commercial potential is limited by the complexity of treating neurodevelopmental disorders and the significant translational gap between mouse proteomic correction and human clinical efficacy.

Market Size 9/10

The research targets Autism Spectrum Disorder (ASD), a massive global market with high unmet medical need and a large patient population. Furthermore, the identification of a specific, conserved molecular mechanism (NO signaling) suggests a potential platform for treating multiple genetic subtypes of ASD.

Tech Readiness 3/10

The research is in the early preclinical stage, focusing on mechanistic proteomic profiling and behavioral validation in murine models. While it identifies a potential therapeutic target (nNOS inhibition), it lacks the pharmacokinetic, safety, or dose-escalation data required for clinical translation.

IP Strength 4/10

The innovation relies on 7-Nitroindazole, a well-known, off-patent small molecule, which limits the ability to claim novel composition of matter. While the proteomic targets identified could potentially support new method-of-use patents, the underlying mechanism is incremental rather than a breakthrough platform technology.

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15. Model-Based Attribution of Biologic Effectiveness in Psoriasis: Drug vs Context Contributions.

26/50

"AI-driven software predicts patient response to biologics, reducing trial-and-error in immunology treatments."

Main researcher: Yuval Ramot · yuval.ramot@mail.huji.ac.il.

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Published in Journal of the American Academy of Dermatology · 2026 Jul 15

WHY NOW

The high cost and variable efficacy of biologic drugs necessitate better patient stratification tools. This precision medicine software offers significant cost savings for healthcare systems and improved outcomes for patients, addressing a key need in immunology.

COMMERCIAL ANGLE

Development of a precision medicine software tool for predicting patient response to biologic therapies, reducing trial-and-error prescribing in immunology.

COMMERCIALISATION METRICS

Novelty 6/10

The approach of using model-based attribution to decouple drug effects from patient context is a sophisticated application of causal inference/ML, but it builds on existing attribution frameworks rather than inventing a new scientific paradigm.

Commercial Potential 6/10

The work suggests a high-value application in precision medicine and companion diagnostics for biologics, but without the abstract, it is unclear if this is a general theoretical framework or a scalable, proprietary tool.

Market Size 8/10

Psoriasis is a high-prevalence chronic condition with a massive, high-value biologics market; tools that optimize drug effectiveness and patient stratification represent a significant commercial opportunity.

Tech Readiness 3/10

The title indicates a model-based theoretical or analytical framework for attributing drug effectiveness, which is characteristic of early-stage computational research (TRL 3) rather than a validated clinical tool or product.

IP Strength 3/10

The title suggests a methodological or diagnostic framework for attribution analysis, which is typically difficult to patent and often categorized as a mathematical method or software tool with low inherent defensibility.

View in Dashboard Open Paper

16. Strategic Template Filtering Accelerates Fragment-Based Peptide Docking. 26/50

"High-efficiency AI algorithm accelerates peptide-protein docking, boosting drug discovery platforms."

Main researcher: Ora Schueler-Furman

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Published in Journal of chemical information and modeling · 2026-06

WHY NOW

Speed and accuracy are paramount in drug discovery. This optimized computational tool significantly reduces the time and cost associated with designing peptide-based therapeutics, making it a valuable asset for licensing to biotech and pharma.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

COMMERCIALISATION METRICS

Novelty 4/10

The work describes an incremental optimization of an existing protocol (PatchMAN) by adding filtering and localization to reduce compute time. It is an efficiency improvement rather than a groundbreaking shift in methodology or a novel scientific discovery.

Commercial Potential 6/10

The tool provides a clear efficiency gain for an existing pipeline, making peptide-protein docking more accessible for drug discovery. However, it is an incremental optimization of a protocol rather than a novel, standalone platform or a direct therapeutic lead.

Market Size 8/10

Peptide-protein interaction modeling is a foundational platform technology for the multi-billion dollar biologics and drug discovery markets. Reducing the computational cost of high-accuracy docking directly enables larger-scale virtual screening and accelerates the development of peptide-based therapeutics.

Tech Readiness 4/10

The work is a computational optimization of an existing software protocol, moving it from a high-cost academic tool to a more efficient one, but it remains in the 'tool development' stage without evidence of integrated drug discovery pipeline application.

IP Strength 4/10

The innovation is an incremental optimization of an existing open-source protocol (PatchMAN) focused on computational efficiency rather than a novel, proprietary algorithm. As a software refinement for fragment filtering, it is likely viewed as an obvious improvement in the field, making it difficult to secure strong, broad patents.

View in Dashboard Open Paper

17. Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor. 26/50

"Novel inhibitor overcomes drug resistance in glioblastoma, significantly boosting chemotherapy efficacy."

Main researcher: Haitham Amal

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Published in Cancer medicine · 2026 Jul 1

WHY NOW

Glioblastoma remains a highly lethal cancer with limited treatment options and frequent resistance to standard chemotherapy. This adjuvant therapy offers a high-value opportunity to extend patient survival and enhance current treatment paradigms.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

COMMERCIALISATION METRICS

Novelty 5/10

While the combination therapy targeting TMZ resistance is clinically relevant, the use of nNOS inhibitors to modulate nitrosative stress in glioblastoma follows a known mechanistic pathway rather than introducing a disruptive or first-in-class biological paradigm.

Commercial Potential 7/10

The study demonstrates a clear combination therapy strategy using an existing small molecule (BA-101) to address the high-value clinical need of TMZ resistance in glioblastoma. While the mechanism is well-validated in preclinical models, the score reflects the transition from an existing compound to a new indication, which involves significant clinical trial risk and complex regulatory pathways for adjuvant therapies.

Market Size 6/10

The target market is a high-unmet-need oncology indication (Glioblastoma), but the total addressable patient population is relatively small and niche compared to broad-spectrum oncology markets. While the combination therapy approach adds value to the existing standard of care, the market size is constrained by the specific prevalence of TMZ-resistant GBM.

Tech Readiness 4/10

The research has progressed beyond basic in vitro studies to include in vivo validation in mouse xenograft models, but it remains in the preclinical stage without evidence of GLP-toxicology or clinical trial data.

IP Strength 4/10

The innovation relies on repurposing an existing small molecule (BA-101) for a new indication, which offers limited composition-of-matter patentability and faces high competition from existing prior art in the nNOS inhibitor space. Defensibility is restricted to method-of-use patents, which are generally easier to circumvent than novel chemical entities.

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